

Learning compositional design rationale from synchronised audio and professional lighting control

"I just want to bring my intent into the machine."

Research gap

Current systems that turn music into light map audio features to looks, predict colour values frame by frame, or learn from recorded control output [1, 2, 3]. They reproduce a style, not the reasons behind a designer's decisions [5].

A professional lighting programme carries structure that spans the whole song. Recurrence, restraint, reveal and the deliberate refusal to react to a beat are design decisions. None of them can be read from the control values the lighting console sends to the luminaires.

Dataset

101 5 1

show records designers production setup

Professional grandMA3 show files, synchronised to each song's audio. Other datasets record the control output to the luminaires, or capture the stage on video [2, 3]. Here the show file itself survives, with its cues, presets, effects and the building blocks each cue refers to.

One shared rig and template keep all shows comparable. The constraint also shapes how designers work, so it is reported as part of the method.

Decomposition

The first part of this work is complete. Decomposing all 101 show files recovers maintained looks, animated layers, accent events, recurrences and how each show develops. Unproven output and unresolved reactions stay marked as uncertain.

Because every layer is explicit, a donor show can already be transferred onto new audio (Fig. 2).

Where this stands

Done. Decomposition of all 101 show files, with reproducibility checks.

Prototype. A first donor-transfer prototype generates editable shows for new songs. Whole-song musical review is ongoing.

Open. The generator architecture. Three candidates, one masked model, a plan-then-realise model and a diffusion cascade, will be compared in a small matched pilot.

Next. Representation round trip, generation for unseen songs and the hierarchy diagnostic.

Proposed representation

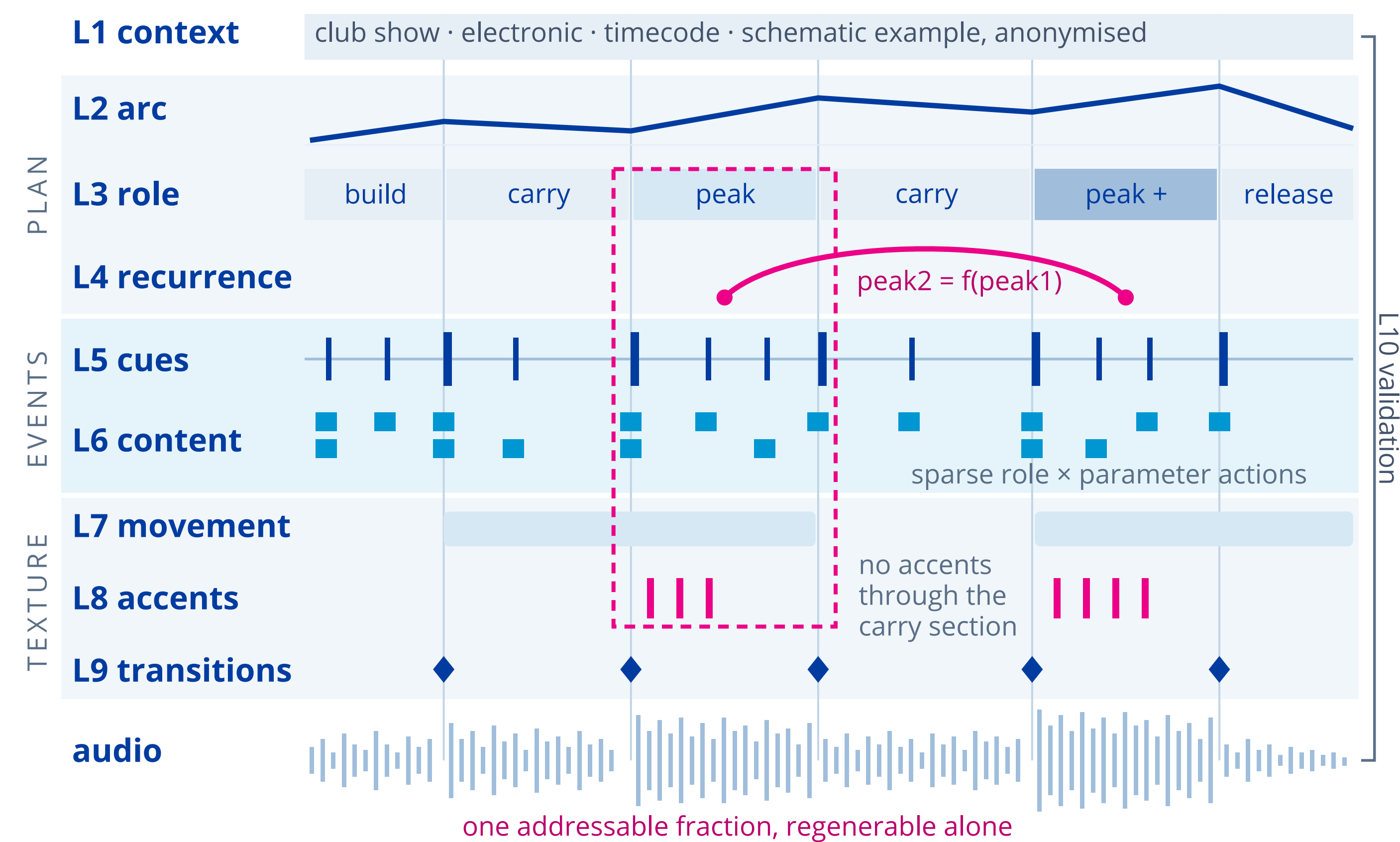


Fig. 1: Proposed layered representation above synchronised audio, shown schematically. Context (L1) and the dramaturgic arc (L2) sit above seven layers in three bands (plan, events, texture), each with its own kind of object. Validation (L10) checks across all of them. The dashed region marks a selected time region. Regenerating it should leave protected content unchanged, which is still to be tested.

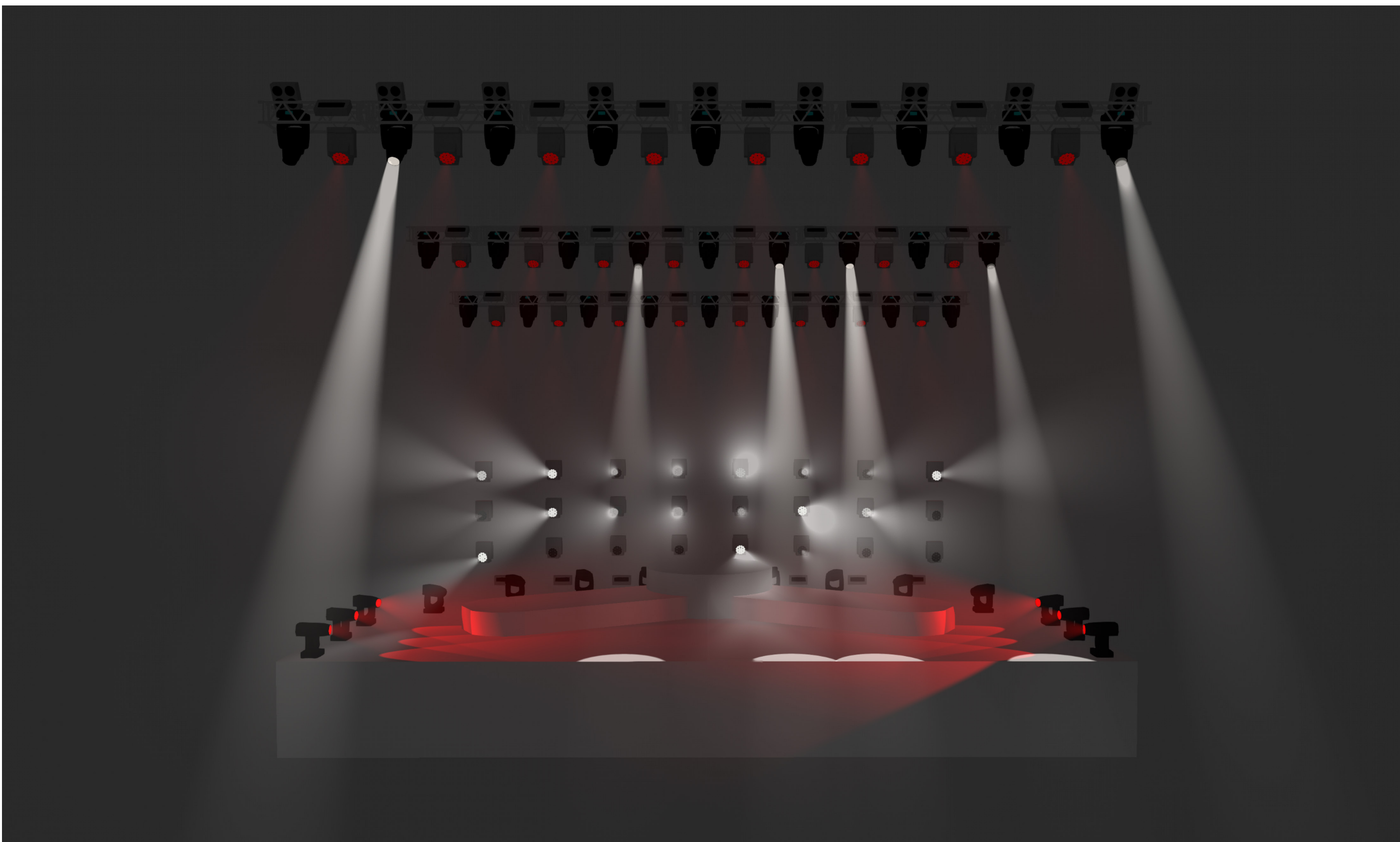


Fig. 2: grandMA3 3D visualiser. A show the current prototype generated for a new song with no authored show in the dataset (Billie Eilish, bad guy). The prototype transfers the structure of a donor show onto the new audio. This only works because the decomposition makes the donor's structure explicit.

Research question

Can a hierarchy-aware audio-conditioned model generate an addressable, layered representation of professional concert-lighting structure and console semantics for unseen songs, within a fixed production environment, such that selected layers or temporal fractions can be regenerated while protected content stays stable?

Four hypotheses follow from this question. Each is tested separately.

H1 Representational adequacy. Authored shows are encoded into the smaller layered representation and rebuilt from it.

H3 Hierarchy utility. Predicting each layer from the layers it depends on beats equally sized flat and collapsed models.

Decomposition in numbers.

The 101 records yield 1,338 sections, 3,164 main cues that change the look, 12,765 accent events and 569 recurrence families within shows. The counts differ by an order of magnitude between layers, so each scale needs its own model part. They are not independent training examples.

H2 Predictive adequacy. The representation is generated from an unseen song's audio, musical structure and context.

H4 Intervention integrity. A selected layer or time region is regenerated while everything declared as protected stays unchanged.

Proposed generation

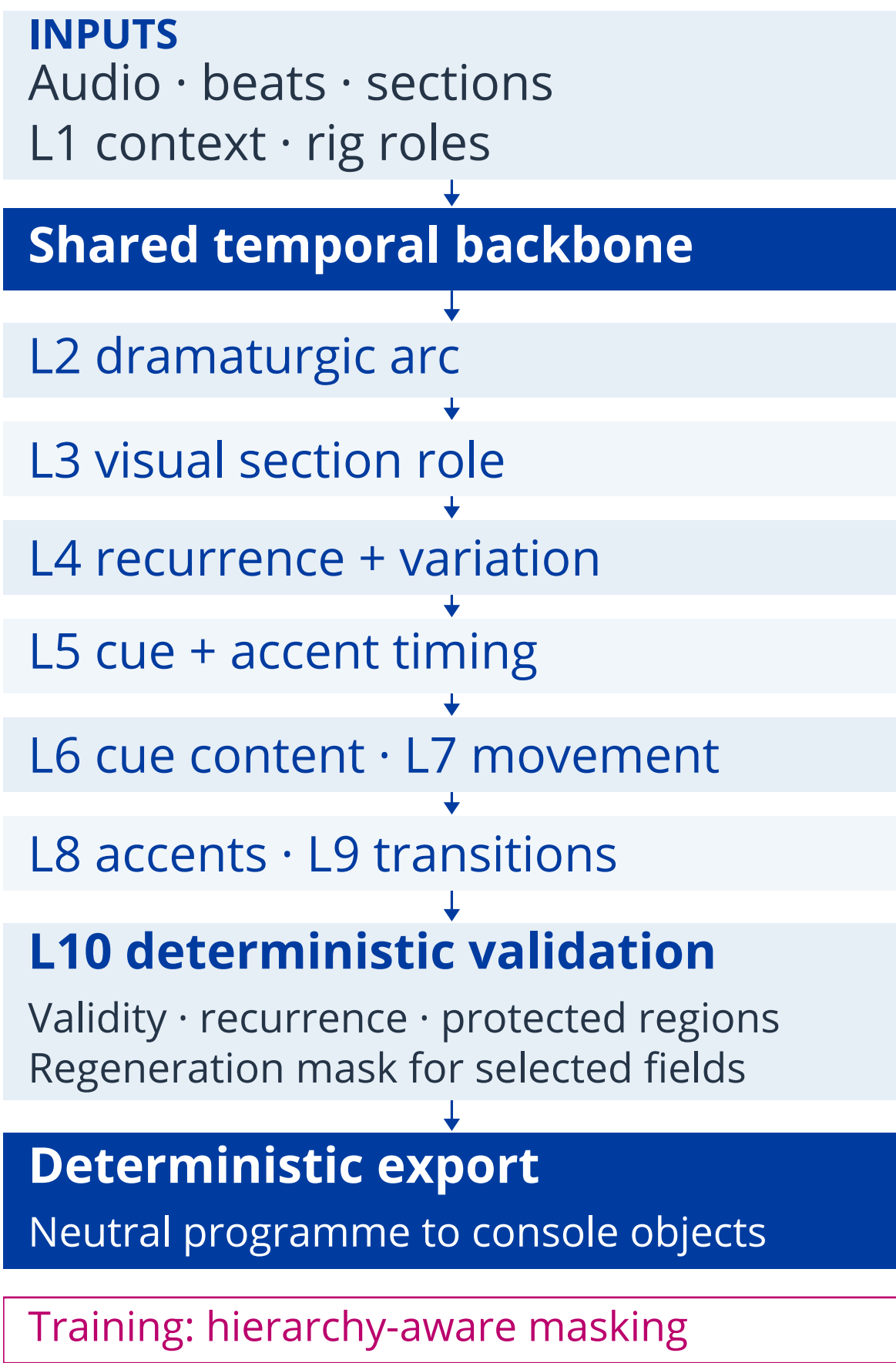


Fig. 3: Proposed generation architecture, following hierarchical music generation [4]. Each layer is predicted from the layers it depends on. One masked objective covers generation and editing. Validation and export follow fixed rules.

Hierarchy diagnostic

Three variants of the chosen model, equal in size, input, training signal and output. They differ only in which layers see which other layers.

Flat. One model predicts every layer at once. No layer sees another layer's prediction, but every layer is still trained.

Collapsed. Two steps. A section plan first, then everything else.

Hierarchical. Each layer is predicted from the layers it depends on.

How it will be judged

Each variant is measured layer by layer, from the arc down to transitions. At every layer we report similarity to the authored show, plausibility against all authored shows, alignment with the musical structure and validity on the console, each separately. Several different shows can be right for the same song, so no single score decides.

Test songs are unseen. No recording of the same musical work is used for training, normalisation or thresholds, and all thresholds are fixed before results are seen.

Not claimed

The designers' actual reasons. Artistic merit. Acceptance and usefulness for practitioners. Generalisation beyond these five designers and this one rig.

Poster, project
& contact

[github.com/twotonetobi/
rq1-lighting-representation](https://github.com/twotonetobi/rq1-lighting-representation)

Work in progress
September 2026

tobias.wursthorn@haw-hamburg.de

